ET Gen AI Hackathon Proposal Submission

AutoTeX AI: The 'Zero-Layout' Generative Publishing Engine

Transforming Raw Content into Broadcast-Quality Documents via Agentic LaTeX

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Problem Statement

While Generative AI has solved content creation (text generation via LLMs, image generation via Diffusion), content presentation remains a manual bottleneck. Media houses, hyper-local publishers, and researchers spend 40% of their time on formatting, layout alignment, and design consistency1 rather than content. Current "AI Writers" output unformatted text blocks, requiring human designers to use complex tools (InDesign, LaTeX) to make them publication-ready.



Motivation

While Generative AI has solved content creation (text generation via LLMs, image generation via Diffusion), content presentation remains a manual bottleneck. Media houses, hyper-local publishers, and researchers spend 40% of their time on formatting, layout alignment, and design consistency rather than on content. Current "AI Writers" output unformatted text blocks, requiring human designers to use complex tools (InDesign, LaTeX) to make them publication-ready.

Student Track

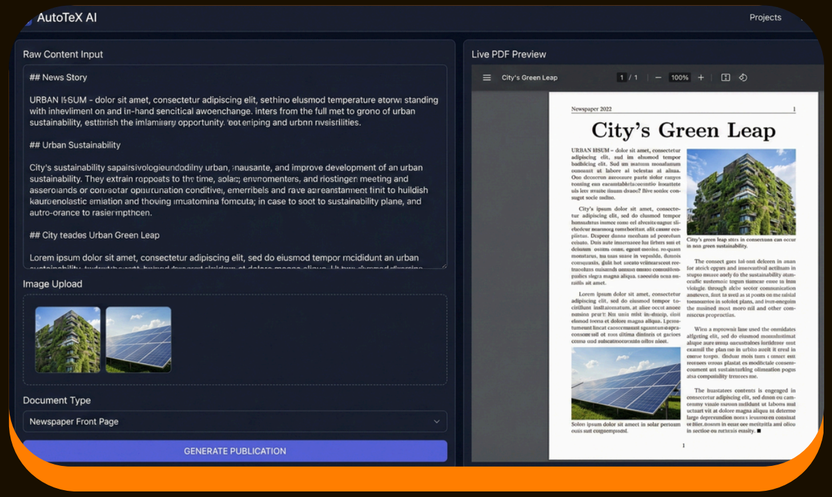
Submission by:- Team Compass Interactives

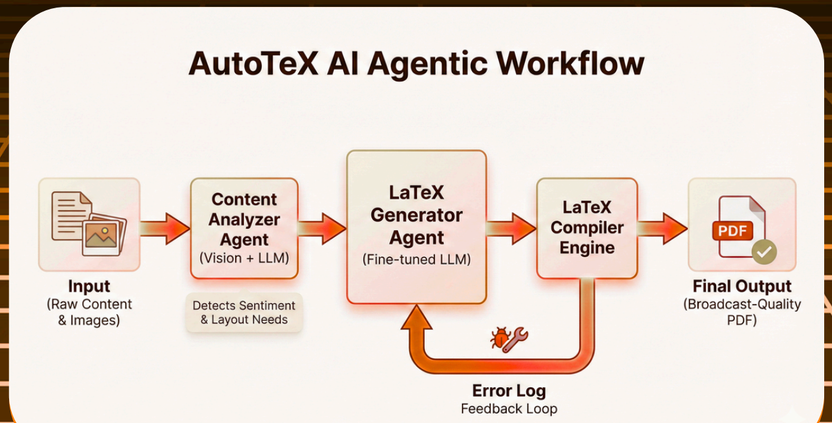
About AutoTex AI

Proposed Method

We utilize an Agentic Workflow rather than a simple prompt-to-text model.

1. Content Analyzer Agent (Vision + LLM): Analyzes input text and images to determine "Document Sentiment" (e.g., Academic requires dense text; Magazine requires large visuals).
2. Layout Orchestrator: Maps content to a dynamic grid system.
3. Code Generator (The Core): Fine-tuned LLM (e.g., Llama 3 or GPT-4o) generates modular LaTeX code.
4. The "Self-Healing" Loop (Novelty): The system attempts to compile the PDF. If the LaTeX engine throws an error, the error log is fed back to the LLM to auto-correct the syntax and re-compile, meanwhile collecting data for further finetuning the LLM.
5. Rendering Layer: Output as PDF/HTML5.





Experiments

- Compilation Success Rate (CSR): Measuring how often the AI fixes its own LaTeX syntax errors (Target: >95%).
- Layout Diversity Score: Using Computer Vision to ensure the AI doesn't just reuse one template but adapts the layout to image sizes.
- Time-to-Publish: Comparing "AutoTeX" vs. "Manual InDesign Workflow" (Expected reduction: 2 hours to 2 minutes).

Dataset/Data Source

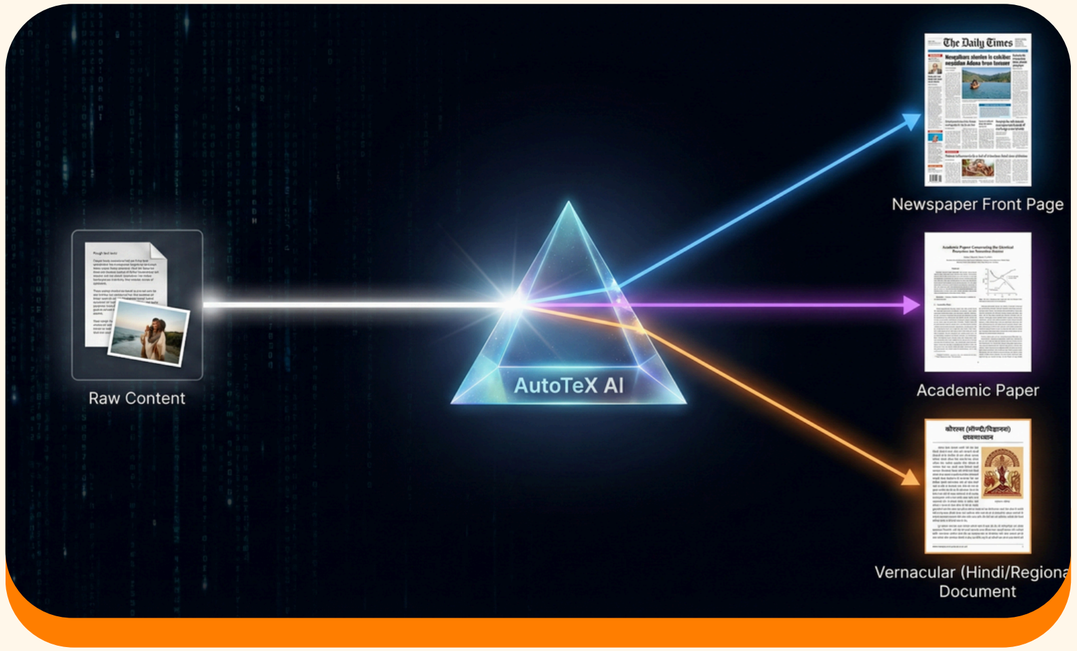
- Layouts: Scraped repository of 5,000+ open-source Overleaf templates and arXiv source files (for structure learning).
- Design Rules: A custom "Design Grammar" dataset mapping sentiment to typography (e.g., "Serious" = Serif fonts, "Tech" = Sans Serif).

Impact

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Efficiency Gap	Reduces "Time-to-Publish" from 4 Hours to 4 Minutes.	Redirects 40% of editorial budget back to reporting.
The ‘ET’ Angle	Standard GenAI struggles with complex Indian typography.	Using LaTeX, we get ready to print Indian Typography.
Future Scope	B2B: Plugin for WordPress/CMS (Sold to Media Houses).	B2C: "Resume & Assignment" subscription for students.

Novelty

- The Iterative Compilation Loop. Most GenAI tools guess code and fail. Ours runs a compiler in the loop to guarantee the output works.
- API-First: can be integrated into any CMS (WordPress/Drupal) as a "Save as PDF" plugin.
- Zero-Shot Style Cloning via Visual Prompting. Unlike traditional editors limited to fixed libraries, AutoTeX uses Vision-LLMs to reverse-engineer layout geometry. Users can upload a screenshot of any publication (e.g., a vintage newspaper), and the system auto-generates the LaTeX class file to clone that exact aesthetic for their new story, enabling infinite, custom layout possibilities without coding.

Application

- Primary Sector: Digital Media & Smart Journalism (ET Hackathon Track)Newsrooms: Instantly converting a breaking news wire feed + 3 images into a formatted PDF front page.
- EdTech: Auto-generating perfectly formatted math/science worksheets from a syllabus prompt.
- Real-World Scenario: A journalist feeds raw notes and a photo of a protest. AutoTeX identifies the sentiment (urgent/serious), selects a "Breaking News" layout, places the image dynamically, writes the LaTeX code, and exports a print-ready PDF in 30 seconds.

Thank You